

MONROE, WI, USA

WMO: 726414

Lat: 42.615N

Lon: 89.591W

Elev: 1085

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04873

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.6	(c) 0.0	(d) -13.0	(e) 2.8	(f) -3.9	(g) -8.4	(h) 3.6	(i) 0.9	(j) 28.5	(k) 26.8	(l) 25.9	(m) 21.5	(n) 10.0	(o) 270	(p) 0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	89.6	72.8	86.2	71.6	83.6	70.8	76.7	83.8	74.9	82.3	73.2	80.5	11.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	135.1	80.4	72.6	125.7	78.8	71.6	121.6	78.1	40.9	84.2	39.1	82.3	37.6	81.4	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	21.2	18.9	-11.9	92.5	7.2	3.0	-17.0	94.6	-21.2	96.4	-25.3	98.0	-30.5	100.2
			-12.4	79.7	7.1	1.5	-17.5	80.7	-21.6	81.6	-25.6	82.4	-30.7	83.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	20.9	23.3	35.9	47.7	59.3	68.9	72.0	70.6	63.8	50.8	38.4	26.2
	DBStd	20.38	11.79	11.35	11.62	9.69	8.83	6.14	5.54	5.21	7.65	9.21	9.93	10.90
	HDD50	3496	901	749	460	154	18	0	0	0	4	104	366	740
	HDD65	6798	1366	1168	904	523	224	30	8	12	112	449	798	1204
	CDD50	2868	0	1	23	87	305	566	682	638	418	129	18	1
	CDD65	696	0	0	2	5	46	147	225	186	76	9	0	0
	CDH74	5620	0	0	12	65	407	1208	1895	1356	599	78	0	0
	CDH80	1580	0	0	0	10	104	343	594	361	156	12	0	0
(14) Wind	WSAvg	8.6	9.9	9.9	9.6	10.4	8.8	7.4	6.6	6.1	7.0	8.8	9.6	9.5
(15) Precipitation	PrecAvg	37.1	1.4	1.5	2.2	3.8	3.9	5.7	4.0	4.3	3.6	2.7	2.3	1.8
	PrecMax	47.1	3.3	3.5	5.6	6.7	8.7	12.8	8.2	14.8	7.4	5.7	6.0	4.6
	PrecMin	27.5	0.3	0.1	0.5	1.2	0.6	1.2	1.2	1.4	0.9	0.5	0.2	0.5
	PrecStd	5.0	0.7	0.9	1.5	1.6	1.8	2.8	1.9	2.9	1.6	1.5	1.5	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	56.9	74.9	81.5	89.7	90.8	95.4	91.1	89.8	81.5	68.0	59.0
		MCWB	44.9	49.3	62.7	62.1	69.1	73.4	73.4	72.3	71.3	66.9	57.6	55.8
	2%	DB	43.4	46.2	63.9	73.5	82.0	87.8	89.1	87.6	84.1	75.1	62.7	49.6
		MCWB	39.4	42.1	56.2	58.3	65.8	71.7	74.2	72.6	70.2	63.5	54.9	45.6
	5%	DB	38.7	41.4	58.9	69.7	78.9	83.9	85.9	83.8	81.1	70.4	57.1	44.6
		MCWB	36.5	37.6	52.1	55.9	63.9	69.9	72.7	71.3	67.7	60.8	52.0	42.4
	10%	DB	36.1	36.9	52.4	63.9	73.4	81.3	82.4	81.3	77.1	65.6	53.8	39.4
		MCWB	34.6	34.7	46.1	53.2	62.4	68.6	71.0	69.6	65.5	57.9	49.1	37.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.0	50.1	63.1	65.6	72.4	77.3	80.0	78.0	75.7	70.9	60.5	57.0
		MCDB	47.2	55.1	71.9	75.1	82.7	86.0	85.8	84.9	83.7	78.1	64.8	59.1
	2%	WB	40.1	42.9	58.9	61.3	69.1	75.2	77.5	75.9	72.7	66.6	56.4	46.4
		MCDB	43.4	46.4	64.2	70.2	77.6	82.8	83.3	82.4	79.8	72.0	60.4	47.3
	5%	WB	36.8	38.3	53.5	58.3	66.9	73.2	75.2	74.0	70.5	62.8	53.3	42.2
		MCDB	38.8	41.2	57.9	67.6	74.8	80.7	83.0	80.4	76.6	67.2	56.8	44.0
	10%	WB	34.2	35.2	46.1	54.3	64.5	71.0	73.3	72.2	68.4	58.8	48.4	37.5
		MCDB	35.7	37.1	50.9	62.5	71.6	78.2	80.9	78.4	74.5	65.2	53.1	39.1
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	14.2	15.6	19.0	19.5	18.5	18.2	17.8	19.3	17.2	14.7	12.2
		MCDBR	15.1	17.3	22.5	26.6	23.4	21.2	20.4	20.7	22.9	22.9	20.5	16.1
		MCWBR	12.7	13.3	14.5	15.2	11.7	10.6	9.4	9.2	10.7	12.7	14.7	13.6
	5% WB	MCDBR	14.8	16.7	20.3	23.3	20.1	18.9	18.6	17.4	18.8	19.9	19.1	15.6
		MCWBR	12.7	13.3	15.1	15.7	12.0	11.1	10.5	10.0	10.7	15.0	15.4	14.2
(40) Clear-Sky Solar Irradiance	taub		0.279	0.301	0.326	0.370	0.397	0.414	0.426	0.405	0.376	0.333	0.294	0.277
	taud		2.387	2.334	2.400	2.298	2.276	2.266	2.230	2.290	2.349	2.472	2.514	2.433
	Ebn at Noon		275	285	292	284	278	273	268	270	270	271	269	266
	Edh at Noon		25	31	33	40	42	42	43	40	35	27	22	22
(44) All-Sky Solar Radiation	RadAvg		549	867	1191	1464	1718	1935	1931	1698	1395	906	590	432
	RadStd		68	101	124	121	116	143	109	109	110	110	50	35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page